WAN Configuration and Communication Between Two LANs

What is LAN?

LAN (Local Area Network) is a network that connects computers and other devices within a small area, such as a home, school, office, or college.

Example: Computers connected to the same switch in a computer lab form a LAN.

What is WAN?

WAN (Wide Area Network) is a network that connects different LANs over a larger area. Routers are commonly used to connect these networks.

Example: In your Packet Tracer project, LAN 1 and LAN 2 are connected through two routers using a WAN link.

How to do set up on cisco packet tracer:-

1)Create 1st LAN:-

a)Take 2 Pc , 1 Switch and 1 router

b)Connect both PC to switch and the switch to router through a wire

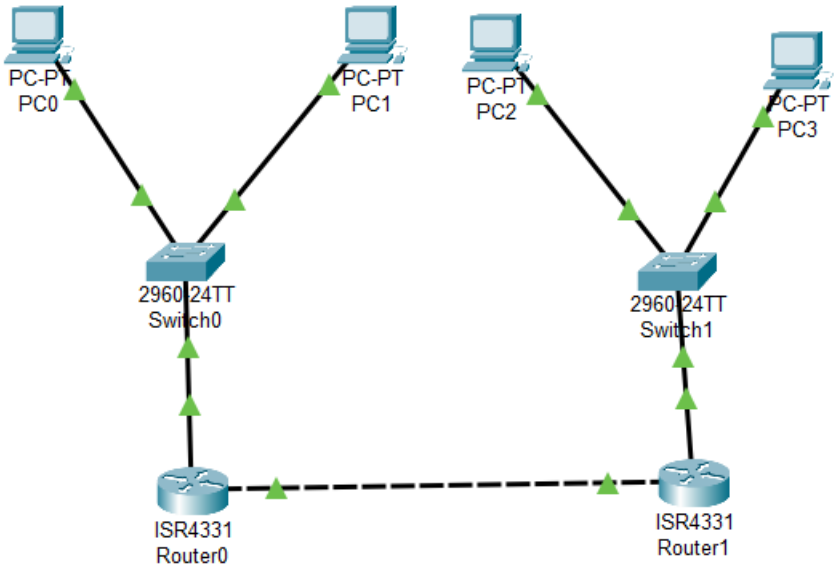
2) Create 2nd LAN:-

a)Take 2 Pc , 1 Switch and 1 router

b)Connect both PC to switch and the switch to router through a wire

3) Connect both LANs Router to each other through the wire

Harshika Bhosale



How to assign IP address to them

1st Lan:-

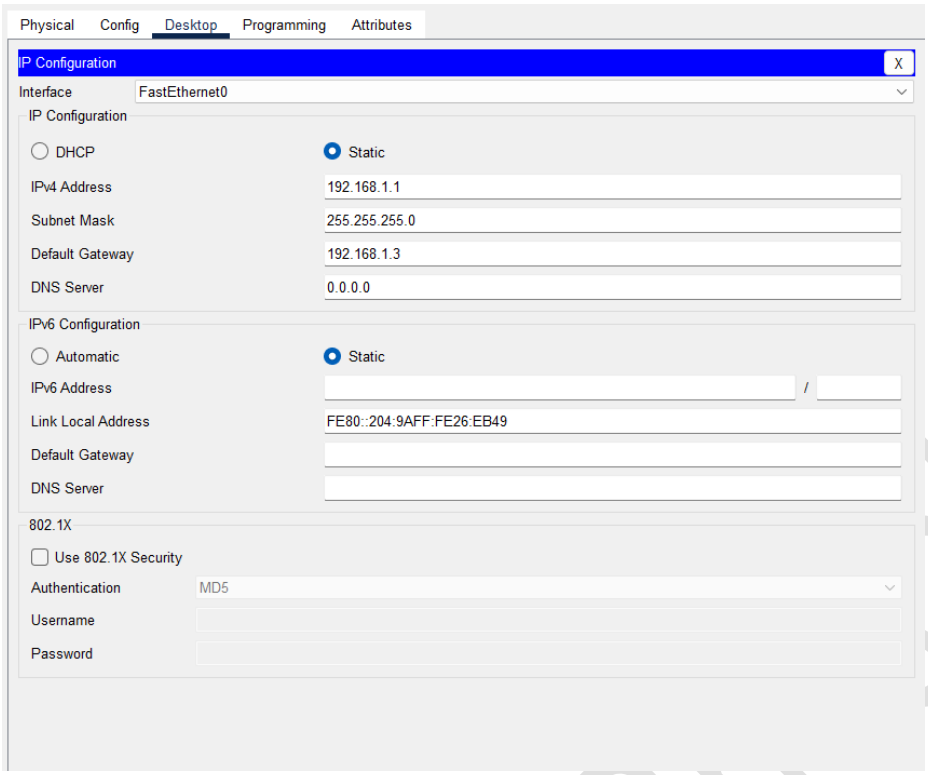
Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.1	255.255.255.0	192.168.1.3
PC1	192.168.1.2	255.255.255.0	192.168.1.3
Router 0	192.168.1.3	255.255.255.0	

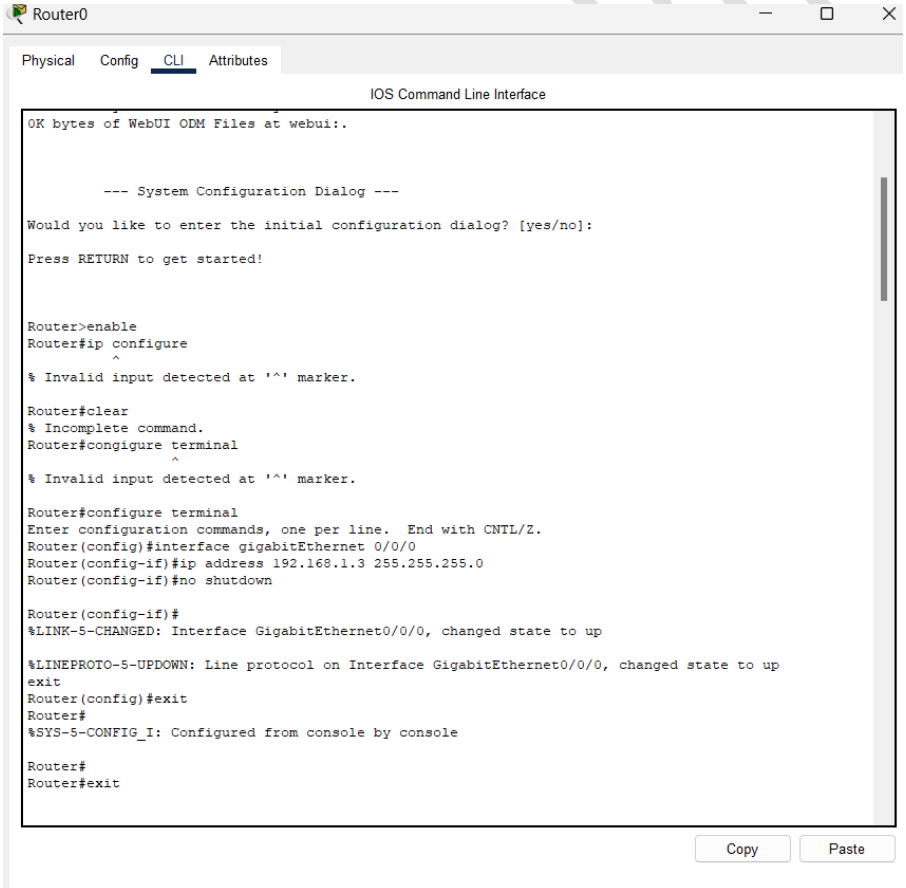
2nd Lan:-

Device	IP Address	Subnet Mask	Default Gateway
PC2	192.168.2.1	255.255.255.0	192.168.2.3
PC3	192.168.2.2	255.255.255.0	192.168.2.3
Router1	192.168.2.3	255.255.255.0	

Routers Wan:-

Device	IP Address	Subnet Mask
Router0 WAN	192.168.3.1	255.255.255.0
Router1 WAN	192.168.3.2	255.255.255.0


Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.1
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.3
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::204:9AFF:FE26:EB49
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
OK bytes of WebUI ODM Files at webui:.
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]:
Press RETURN to get started!
Router>enable
Router#ip configure
^
% Invalid input detected at '^' marker.
Router#clear
% Incomplete command.
Router#congigure terminal
^
% Invalid input detected at '^' marker.
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0/0
Router(config-if)#ip address 192.168.1.3 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#exit
Copy
Paste

⚙️ Configuration

1. Router0 LAN Configuration

The LAN interface of Router0 was configured with:

interface gigabitEthernet 0/0/0

ip address 192.168.1.3 255.255.255.0

no shutdown

2. Router1 LAN Configuration

The LAN interface of Router1 was configured with:

interface gigabitEthernet 0/0/0

ip address 192.168.2.3 255.255.255.0

no shutdown

3. WAN Configuration

Router0 and Router1 were connected using a WAN/serial link.

Router0

interface serial 0/0/1

ip address 192.168.3.1 255.255.255.0

no shutdown

Router1

interface serial 0/0/1

ip address 192.168.3.2 255.255.255.0

no shutdown

The exact serial interface name may differ depending on the router/module used in Packet Tracer.

Harshika Bhosale

Static Routing

Static routes were configured so that each router knew how to reach the remote LAN.

Router0

ip route 192.168.2.0 255.255.255.0 192.168.3.2

This route tells Router0 to forward traffic for the 192.168.2.0/24 network to Router1.

Router1

ip route 192.168.1.0 255.255.255.0 192.168.3.1

This route tells Router1 to forward traffic for the 192.168.1.0/24 network to Router0.

Connectivity:

1)First check 1st Lan connectivity

a)Check if PC can connect each other

b)Check if PC can connect there router

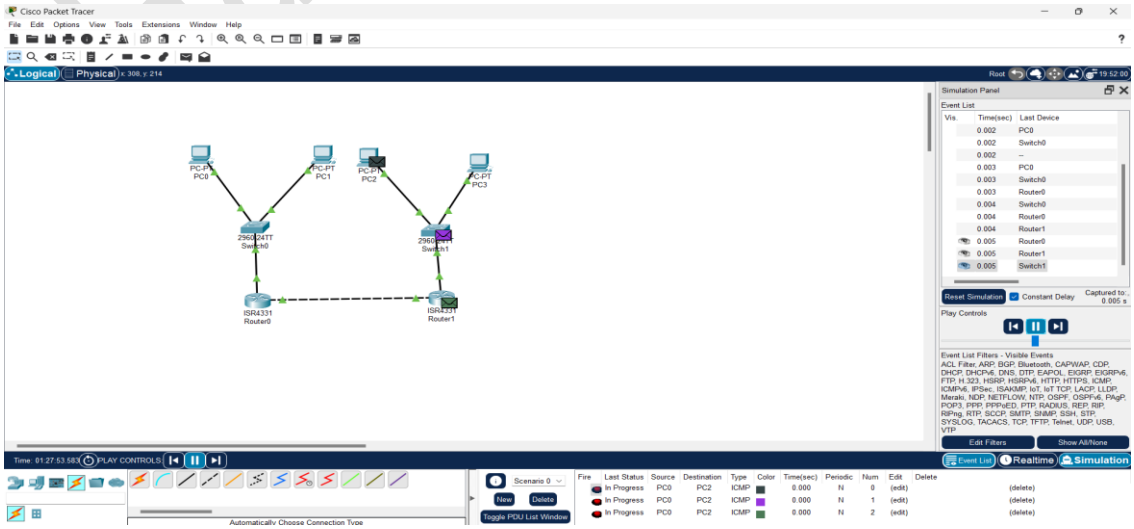
2)First check 2nd Lan connectivity

a)Check if PC can connect each other

b)Check if PC can connect there router

3)Check if routers and connect each other

4)Check if PC from 1st LAN can connect 2nd LANs PC



Harshika Bhosale

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